Google Cloud Cybersecurity Certificate

Module 1

- Identifying and protecting against threats in the cloud
- Detecting, responding, and recovering from cloud cybersecurity attacks

Overview of Cloud Infrastructure

The cloud is really important because it's a trajectory of where computation is going. " On prem" is an information technology infrastructure that is physically located in an organization's own data center or office. This means the organization manages their own servers, networks, and storage to run and store data.A data center is a physical building that stores servers, computer systems, and associated components creating a centralized location for vast amounts of data.

A zone is the collective number of data centers in an area. A zone may contain one data center or multiple data centers. And a region is a group of zones. Google Cloud refers to the grouping of zones and regions as failure domains.A failure domain is a resource that can fail without impacting the availability of data.Failure domains can include zones and regions where data is replicated to improve resiliency.

Latency - the time it takes for data to travel from one location to another. Can be from the user's device to the cloud server, or between cloud servers.
Digital transformation - when an organization modernizes their applications, services and customer relationships by using new technologies.

Cloud Deployment Models

- Public: a cloud model that delivers computing, storage and network resources through the internet, allowing users to share on-demand resources. A multi-tenant environment is an environment in which cloud infrastructure and resources are shared among users.
- Private: a cloud model in which all cloud resources are dedicated to a single user or organization and are created, managed, and owned within on-premises data centers. A single tenant environment in which cloud infrastructure and resources are dedicated to a single user. This model is useful for companies with compliance and/or regulatory considerations.
- Hybrid: a cloud model that combines public and private models so organizations can enjoy both cloud services and the control features of on-premises cloud models. Allows users to add a public cloud infrastructure to an on-premise network.multicloud is a strategy of using more than one cloud service provider.

Cloud Benefits

- Time to market
- Scalability
- Cost savings - follows an opex model (pay only for those resources you use)
- Better collaboration
- Better security ⭐

> Redundancy is the practice of having the same data in different areas and zones

Cloud Limitations

- Continuous internet connection
- Shift in infrastructure control
- Migrating existing systems to cloud

A failure domain is a resource that can fail without impacting the availability of data. In Google Cloud, groupings of zones and regions are called failure domains. A zone is the collective number of data centers in an area.

Ephemerality and cloud resources

Cloud computing - the practice of using on-demand computing resources as services hosted over the internet

Instance - a server resource that runs workloads in the cloud

Ephemerality - the concept that things only exist for a short amount of time, aka change in the cloud is constant

Cloud resources

- **Compute**: computation performed by a physical computer in a remote environment
- **Cloud data storage**: a solution that enables organizations to keep, access and maintain digital data on off-site, cloud based storage devices.

Virtualization and virtual machines

Virtualization: technology that creates virtual versions of physical infrastructure, such as servers, storage, and networks.

Hypervisor: the abstraction layer that sits between the physical computer and the virtual machine.

Types of Hypervisors

- **Type 1: bare metal**
  > interact with OS systems directly
- **Type 2: hosted**
  > uses the computers existing OS and runs as an application
  > easy to host and use, but needs to take care of security

Advantages of VMs

- Portability
- Scalability
- Testing environments

Serverless computing

Backend as a service (BaaS)

Serverless computing is delivered in two main cloud service models: backend as a service (BaaS) and function as a service (FaaS).

BaaS is a form of serverless computing where the CSP manages all aspects of the backend infrastructure. This includes servers, containers, and virtual machines. Developers use BaaS to speed the creation of web applications. With BaaS, developers can focus on writing the front end code, which is the code that builds the user interface. Organizations have access to other services, like databases, file storage, and authentication services that can be native or third party to the platform.

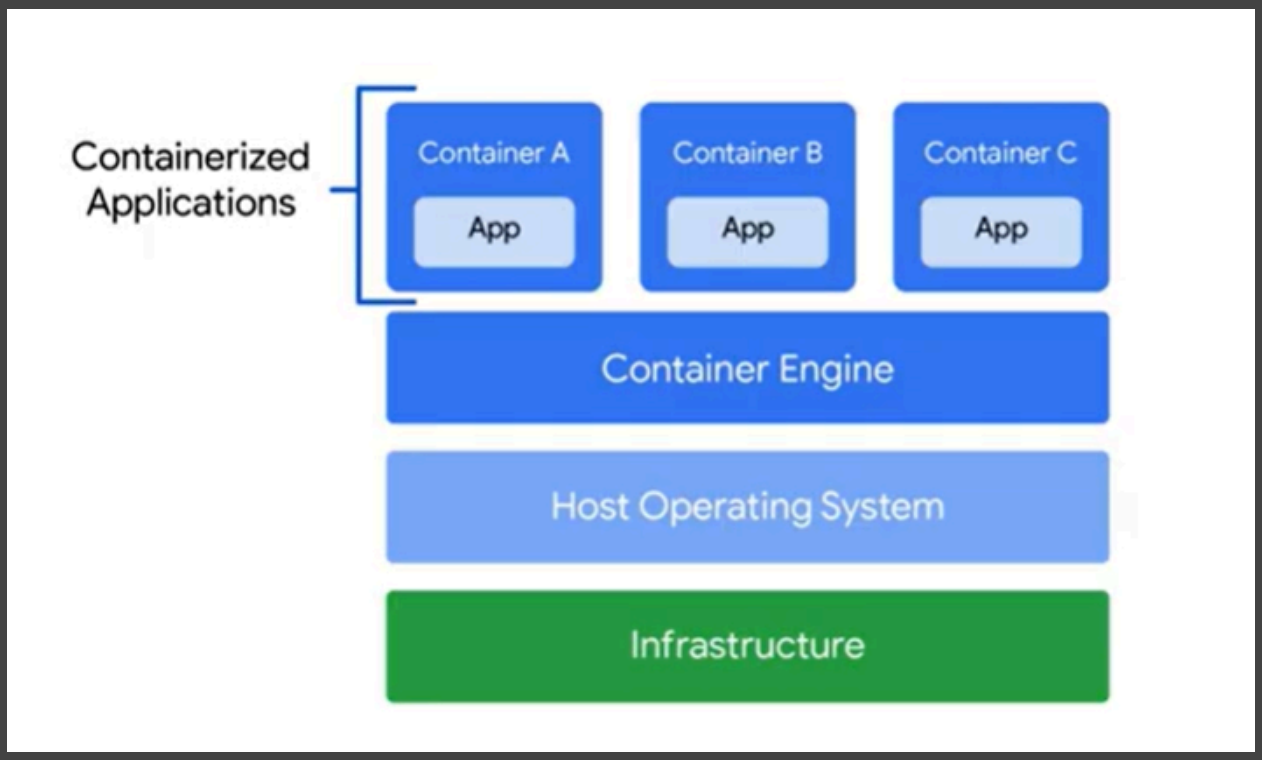
Function as a service (FaaS)

FaaS is a form of serverless computing that runs functions. A function is a small piece of code. Functions are ephemeral, meaning they only exist for a short period of time. Developers can use their choice of programming language to create functions, which makes adopting serverless computing more convenient.

For example, companies can use functions to better engage their customers. Here's how it works: A company chooses an event, like a customer signing up to join their website. Once the customer submits their sign-up request, they receive an email welcoming them to the website.

Containerization

Container: a software package that holds only the components necessary to execute a particular application. Consists of the code and dependencies necessary to run the application. Developers use things called **container registries** to upload and download containers.



Container key benefits

- Portability: run software quickly across computing environments, without a need for a separate OS
- Immutability: an object cannot be changed after it is created and assigned a value
- Responsibility separation:

Quiz

Computation performed by a physical computer in a remote environment is called compute. For example, memory, CPU, and storage are all compute components.

The company should choose Type 1, because it replaces the operating system of the underlying computer, making it more secure. Type 1, or bare metal hypervisors interact with the underlying computer's components directly; they do not run inside of an operating system. Having this direct access makes Type 1 more streamlined and secure.

Packaging software in containers allows for portability, meaning it enables users to quickly run software applications from one computing environment to another. This also uses less memory and fewer resources, because the operating system itself is not included in the container.

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Virtualization technology creates a virtual version of physical infrastructure, like servers, storage, and networks. These virtual machines, or VMs then simulate a physical computer.

Cloud storage benefits

Cloud data storage: a solution that enables organizations to keep, access, and maintain digital data on off-site, cloud based storage devices.

- Scalability
- Redundancy

- Cost savings
- Scalability
- Redundancy
- Security

Types of storage in the cloud

There are a lot of different ways to store data in the cloud, because each storage type has a required datatype

1. Structured data: data organized in a certain format like rows and columns
2. Unstructured data: data that is not organized in an easily identifiable ways

Types of cloud computing storage:

1. **File**: hierarchy of files and folders. For smaller volumes of data with a few people accessing it. Starts at home dir. Oldest and most widely used data system. Used indata repos.
2. **Object**: hold unstructured data in a bucket. Each bucket is assigned a unique way. Can be public or private
   BUCKET: a virtual container that hold objects
3. **Block**: data is split into smaller block each stored in different places for flexibility and efficiency.

Hot and cold data

In storage, there are two ways to describe data and its availability: hot or cold. Hot data is data that users access frequently. For example, the data used in mobile apps is hot because users have instant access to their data.

In contrast, cold data is infrequently or rarely accessed. An example of cold data is data used for record keeping. You might need to keep this kind of data available, but it doesn't need to be updated or accessed often.

Cloud storage

Standard

The standard storage class is best for hot data, or data that you need to access frequently, and for short periods of time. Data stored in this class is highly available and retrievable in milliseconds. Content used for websites, gaming platforms, or streaming videos are all examples of data that fits the standard storage class.

Another type of data that fits well in the standard storage class is medical data. For example, if someone is rushed to the hospital for a broken leg, a doctor will collect data for their medical records, like a description of the injury and x-rays. If the patient needs surgery and physical therapy, it would be best to store their medical data in the standard class, so that it can be accessed quickly.

Nearline

Nearline storage is best for data that you only need to access up to once a month. Nearline is more cost effective than standard storage, and a good option for backing up your data.

In the case of the patient with the broken leg, once their injuries start to heal, they may move to monthly check-ups. With fewer hospital visits, their records will only need to be accessed once a month. So, their data can be moved into nearline storage.

Coldline

People who only need to access their data once every 90 days, or once a quarter, can use coldline storage. Coldline storage is very cost effective because the data being stored is at rest for long periods of time.

For example, when the patient's leg is almost healed, they'll likely only need to see a doctor once every few months. So, their data can be transferred to coldline storage because doctors no longer need frequent access to their records.

Archival

Finally, archival storage is best for archiving or backing up data for disaster recovery purposes. This includes data that is accessed very infrequently, or once a year. Since there's not as high of a demand for availability, archival storage is also very cost-effective.

When the patient's leg is completely healed, and they no longer need check-ups, their data can be moved to archival storage. This means the patient's data is still available for retrieval, but will cost the hospital less money to store.